

STA 35C: Statistical Data Science III

Lecture 25: Hierarchical Clustering

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Announcement

Final exam: Fri, June 5 (1:00 pm–3:00 pm) in Wellman Hall 6 (=classroom)

- **Cumulative:** Lecture 1–Lecture 26
- **Be on time:** The exam starts at 1:00 pm and ends at 3:00 pm sharp
- **Three hand-written cheat sheets allowed:** Letter-size (8.5"×11"), double-sided, brief formulas/notes
- **Calculator:** A simple (non-graphing) scientific calculator is allowed
- **No other materials** beyond the cheat sheets (no textbooks, etc.)
- **SDC accommodations:** Confirm scheduling with AES online as soon as possible

Homework 7 is due tomorrow (Tue, June 2, 11:59 pm)

Course evaluation: Please share your feedback comments by Thu, June 4

Agenda

Last time: *K*-means clustering

- Clustering problem: group observations using only features X
- *K*-means objective: minimize within-cluster variation around centroids
- Practical issues: choosing K , scaling, initialization, local optima

Today: Hierarchical clustering

- Dendrograms and nested clusters
- Agglomerative hierarchical clustering algorithm
- Linkage choices: single, complete, and average
- Worked example and interpretation
- Comparison with *K*-means

Quick review: Clustering problem

Setup:

- **Data:** $\mathcal{X} = \{X_1, \dots, X_n\} \subset \mathbb{R}^p$
- **Goal:** Partition observations into clusters so that
 - observations in the same cluster are similar,
 - observations in different clusters are dissimilar.

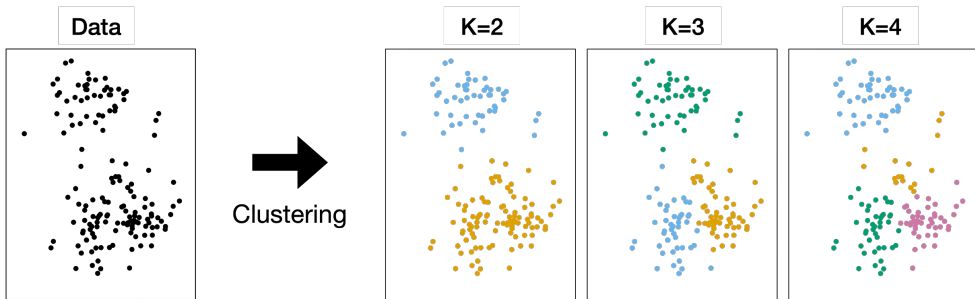


Figure: Given observations (**left**), clustering partitions them into groups (**right**).

Quick review: K -means clustering

K -means: choose K in advance and partition observations into K clusters.

For clusters $C_1, \dots, C_K$, define the centroid

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$

The K -means objective is

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|^2.$$

Algorithm:

1. Initialize cluster labels or centroids.
2. Update centroids.
3. Reassign each observation to the nearest centroid.
4. Repeat until assignments stop changing.

Limitation: K -means must pre-specify K , and the result can depend on initialization.

K-means clustering: Illustration of the algorithm iterations

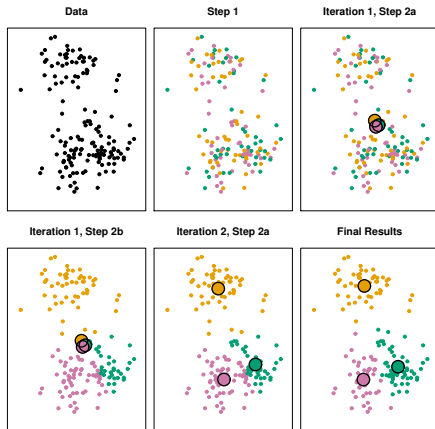


Figure: An example of the K-means with $K = 3$ over two iterations. Each iteration updates centroids (colored disks) and reassigns points [JWHT21, Figure 12.8].

Steps:

- 1 **Initialize** random cluster labels
- 2 **Iterate:**
 - (a) Update cluster centroids
 - (b) Reassign points to nearest centroid

Remarks:

- Each iteration reduces the objective
- Final solution depends on initialization

K-means clustering: Local optima and multiple runs



Key points:

- K-means can converge to a suboptimal (local) solution
- Different initial cluster assignments can yield different final partitions
- Usually, re-run with multiple random starts and pick the best (lowest objective)

Figure: K-means with $K = 3$ repeated six times on the same data, each with a different random initial assignment. Above each plot is the final objective. Multiple local optima are found; the best has objective=235.8 [JWHT21, Figure 12.9].

Hierarchical clustering: Main concept

Motivation: Avoid choosing the number of clusters K in advance

- Instead, build a *dendrogram* that captures how data points “merge” (or “split”) at all levels of (dis)similarity to obtain a hierarchy of nested clusters

Agglomerative (bottom-up) approach:

- 1 Start with n clusters, each containing one observation
- 2 Repeatedly *merge* the two most similar clusters until only one cluster remains
- 3 Record the (dis)similarity at each merge to build a **dendrogram**

Clustering after building a dendrogram:

- “Cut” the dendrogram at a chosen height to produce clusters
- Advantage: A single dendrogram can yield clustering into many different K clusters, depending on where we cut

Dendrograms & cutting for clusters

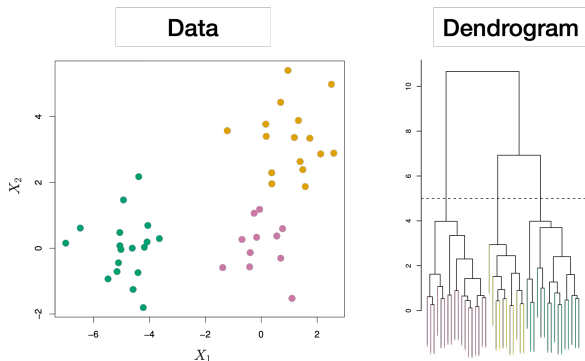


Figure: Left: A synthetic dataset (45 points) in 2D. Right: Its dendrogram, cut at height 5 (dashed line) yielding three clusters (colored). Colors are for display only, not used in clustering [JWHT21, adapted from Figs. 12.10 & 12.11]

Reading a dendrogram:

- Vertical axis = (dis)similarity at which merges occur
- Lower “merge height” = more similar clusters
- Horizontal spacing is not meaningful for distance

Obtaining clusters:

- “Cut” at a chosen height
- The branches below that cut form the clusters
- The method is “*hierarchical*” as lower cuts nest within higher cuts

Interpreting a dendrogram requires care!

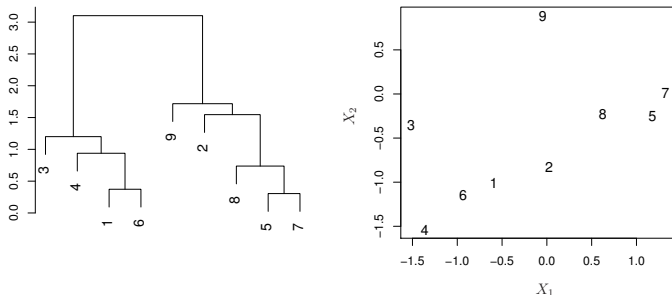


Figure: An illustration of how to interpret a dendrogram with nine observations in 2D. Though points 9 and 2 appear horizontally close, they actually fuse at a higher height than 9 with $\{8,5,7\}$, so 9 is no more similar to 2 than it is to $\{8,5,7\}$ [JWHT21, Figure 12.12].

Note:

- Do *not* interpret horizontal spacing between leaves as distance
- Only merge height indicates dissimilarity
- Dendrogram leaves can often be rotated without changing the clustering

Pop-up quiz: Reading a dendrogram

Suppose a dendrogram has the following merges:

A and B merge at height 1, C and D merge at height 2,

and then $\{A, B\}$ merges with $\{C, D\}$ at height 7.

Question: If we cut the dendrogram at height 3, what clusters do we obtain?

- A) $\{A\}, \{B\}, \{C\}, \{D\}$
- B) $\{A, B\}, \{C\}, \{D\}$
- C) $\{A, B\}, \{C, D\}$
- D) Cannot be determined because the horizontal order is missing

Answer: C. A cut at height 3 is above the merges at heights 1 and 2, but below the merge at height 7. Thus we get two clusters: $\{A, B\}$ and $\{C, D\}$.

Agglomerative hierarchical clustering algorithm

Agglomerative hierarchical clustering algorithm

- 1 **Initialize:** Begin with each observation in its own cluster. Compute **pairwise cluster dissimilarities** (e.g., Euclidean distance).
- 2 **For** $i = n, n - 1, \dots, 2$:
 - (a) Examine all **pairwise inter-cluster dissimilarities (linkage)** among the i clusters and merge the two closest clusters. Record the dissimilarity of that merge as the “height” in the dendrogram.
 - (b) Recompute pairwise distances between the new cluster and all others. Repeat until one cluster remains.

Central question: How do we measure distance between two clusters?

- For individual points, Euclidean distance is common.
- For clusters with multiple points, we need a **linkage rule**.

Hierarchical clustering: Visual illustration of linkage

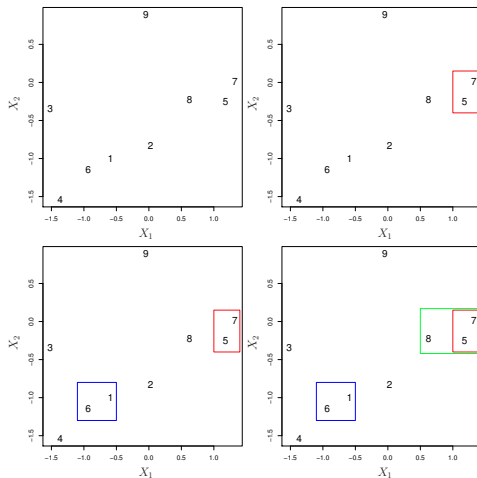


Figure: An illustration of first few steps of hierarchical clustering with complete linkage. **Top Left:** each point is its own cluster. **Top Right:** clusters $\{5\}$ and $\{7\}$ merge first. **Bottom Left:** next, $\{6\}$ and $\{1\}$ merge. **Bottom Right:** now $\{8\}$ merges with the cluster $\{5, 7\}$. Merges occur at heights = pairwise distances [JWHT21, Figure 12.13].

Linkage: Measuring distance between clusters

Let A and B be two clusters.

Single linkage:

$$d(A, B) = \min_{x \in A, y \in B} \|x - y\|.$$

- Distance between closest pair of points.

Complete linkage:

$$d(A, B) = \max_{x \in A, y \in B} \|x - y\|.$$

- Distance between farthest pair of points.

Average linkage:

$$d(A, B) = \frac{1}{|A||B|} \sum_{x \in A} \sum_{y \in B} \|x - y\|.$$

- Average distance across all cross-cluster pairs.

In R: `hclust(..., method = "single"/"complete"/"average")`

Visual comparison of linkage choices

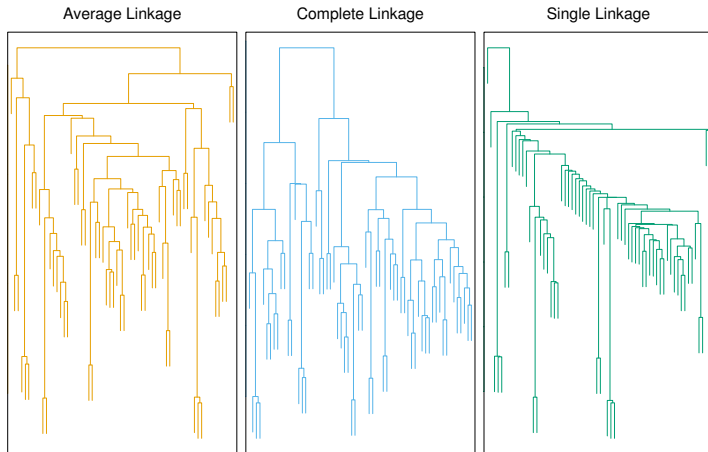


Figure: Comparison of single, average, and complete linkage on the same data. Note that single linkage can produce long “chains,” while complete yields more balanced clusters [JWHT21, Figure 12.14].

Example: Comparing complete vs. single linkage (1/2)

Example

Data set: Let

$$X = \{A = (-3, 2), B = (-1, 3), C = (1, 0), D = (4, -3)\} \subset \mathbb{R}^2.$$

We label these four points $\{A, B, C, D\}$, and first compute all $\binom{4}{2} = 6$ pairwise distances:

$$\begin{aligned}\|A - B\| &= \sqrt{5}, & \|A - C\| &= \sqrt{20}, & \|A - D\| &= \sqrt{74}, \\ \|B - C\| &= \sqrt{13}, & \|B - D\| &= \sqrt{61}, & \|C - D\| &= \sqrt{18}.\end{aligned}$$

Numerically, $\sqrt{5} \approx 2.236$, $\sqrt{20} \approx 4.472$, $\sqrt{74} \approx 8.602$, $\sqrt{13} \approx 3.606$, $\sqrt{61} \approx 7.810$, $\sqrt{18} \approx 4.243$.

Step 1: First merge. The smallest pairwise distance is

$$\|A - B\| = \sqrt{5} \approx 2.236.$$

Hence, both *complete* and *single* linkage begin by merging $\{A\}$ with $\{B\}$, forming a new cluster

$$U = \{A, B\}, \quad \text{so we now have clusters } U, \{C\}, \{D\}.$$

Example: Comparing complete vs. single linkage (2/2)

Example

Step 2: Second Merge. Now we have three branches: $U = \{A, B\}, \{C\}, \{D\}$. We compute their pairwise distances using complete and single linkage, respectively.

Complete linkage merges $\{C\}$ with $\{D\}$ next because

$$\begin{aligned}\text{dist}(U, \{C\}) &= \max\{\|A - C\|, \|B - C\|\} = \max\{\sqrt{20}, \sqrt{13}\} = \sqrt{20} \approx 4.472, \\ \text{dist}(U, \{D\}) &= \max\{\|A - D\|, \|B - D\|\} = \max\{\sqrt{74}, \sqrt{61}\} = \sqrt{74} \approx 8.602, \\ \text{dist}(\{C\}, \{D\}) &= \sqrt{18} \approx 4.243 \text{ (smallest)}.\end{aligned}$$

Single linkage merges $\{A, B\}$ with $\{C\}$ second because

$$\begin{aligned}\text{dist}(U, \{C\}) &= \min\{\|A - C\|, \|B - C\|\} = \min\{\sqrt{20}, \sqrt{13}\} = \sqrt{13} \approx 3.606 \text{ (smallest)}, \\ \text{dist}(U, \{D\}) &= \min\{\|A - D\|, \|B - D\|\} = \min\{\sqrt{74}, \sqrt{61}\} = \sqrt{61} \approx 7.810, \\ \text{dist}(\{C\}, \{D\}) &= \sqrt{18} \approx 4.243.\end{aligned}$$

This example illustrates how different linkages can yield different merges.

Pop-up quiz: Linkage

Suppose $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$, with cross-cluster distances

	b_1	b_2
a_1	2	8
a_2	3	7

Question: What are the single, complete, and average linkage distances between A and B ?

- A) Single = 2, complete = 8, average = 5
- B) Single = 8, complete = 2, average = 5
- C) Single = 2, complete = 7, average = 4
- D) All three are 5

Answer: A. Single linkage uses the minimum distance 2; complete linkage uses the maximum distance 8; average linkage uses $(2 + 8 + 3 + 7)/4 = 5$.

Choosing clusters from a dendrogram

After building the dendrogram, we still choose a final clustering.

Common choices:

- **Cut by height:** choose a dissimilarity threshold.
- **Cut by number of clusters:** choose K , then cut to get K clusters.
- **Use domain knowledge:** choose clusters that are interpretable and meaningful.

Important: Hierarchical clustering avoids choosing K before fitting, but we still choose how to cut the tree afterward.

Practical advice:

- Try multiple linkage rules.
- Check sensitivity to scaling and distance choice.
- Avoid over-interpreting small visual differences in the dendrogram.

Hierarchical clustering: Strengths and limitations

Strengths:

- Does not require choosing K before fitting.
- Produces a dendrogram showing cluster structure at multiple resolutions.
- Can reveal nested structure.

Limitations:

- Greedy merges: once clusters are merged, they cannot be unmerged.
- Sensitive to distance metric, scaling, and linkage choice.
- Can be computationally expensive for large n .

Wrap-up: Clustering summary

Clustering:

- *Goal*: Partition a dataset (no response labels) into subgroups of “similar” observations
- *Unsupervised*: Typically used for exploratory analysis or hypothesis generation
- No single “correct” distance or method; different choices lead to different clusterings

K -means vs. hierarchical clustering:

K-means	Hierarchical
- Partition data into K clusters - Minimizes within-cluster variation	- Builds a <i>dendrogram</i> from bottom-up - Cut at a certain height to obtain clusters
- Simple, computationally fast - Easy-to-interpret “centroids” for each cluster	- No need to specify K in advance - One dendrogram can yield many clusterings
- Must pre-specify K - Local search can yield suboptimal solutions	- Greedy merges rely on linkage choice - Nested clusters may be less optimal

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.